Foundations of Full House Methodology

We now motivate the structure of the Full House Model. Let N_i denote the size of population i. We will suppose the every individual $k = 1, \ldots, N_i$ has an underlying latent aptitude $X_{i,1}, \ldots, X_{i,N_i} \overset{\text{iid}}{\sim} F_X$. We denote $X_{i,(j)}$ as the jth ordered latent aptitude in population i. We define $g_i(\cdot, \cdot)$ as the system inclusion function, where $g_i(X_{i,j}, \mathbf{X}_{i,-j}) = 1$ indicates that individual j is included in the ith system, and $g_i(X_{i,j}, \mathbf{X}_{i,-j}) = 0$ indicates that individual j is not included in the ith system. Let $\mathbf{X}_{i,-j}$ be the vector of all latent aptitudes not including the component j.

We now suppose that an individual j included in system i has an observed statistic $Y_{i,j}$ where we suppose that $Y_{i,1}, \ldots, Y_{i,n_i} \overset{iid}{\sim} F_{Y_i}$, F_{Y_i} is continuous, and $n_i < N_i$ is the number of individuals included in system i. We denote $Y_{i,(j)}$ as the jth ordered statistic for system i.

The key to the Full House Model is connecting latent aptitudes X with observed statistics Y. To achieve this we assume that we can form pairs $(Y_{i,(j)}, X_{i,(r(N_i,n_i,j))})$, $j = 1, \ldots, n_i$, where $r(N_i, n_i, k)$ is the index function that connects this latent aptitude to the kth components in the system i. The function $r(N_i, n_i, k)$ determines the relationship between the components in the system and the selected latent aptitudes in the population. As an example, $r(N_i, n_i, k) = N_i - n_i + k$, then the highest observed statistics $Y_{i,j}$ in system i will be assumed to have the highest latent aptitudes in population i.

We now demonstrate how the Full House Model works in a parametric setting. Consider the pair $(Y_{i,(j)}, X_{i,(r(N_i,n_i,j))})$ and suppose that the distribution corresponding to $Y_{i,j}$ from the ith system is known to belong to a continuous parametric family indexed by unknown parameter θ_i, and let $F_{Y_i}(\cdot \mid \theta_i)$ be a parametric CDF with parameter $\theta_i \in \mathbb{R}^{p_{Y_i}}$. We can estimate θ_i with $\hat{\theta}_i$ and plug the estimator into the CDF $F_Y(\cdot \mid \hat{\theta}_i)$.

In order to connect the $Y_{i,(j)}$ and $X_{i,(r(N_i,n_i,j))}$ and obtain an estimate of the underlying talents, we will make use of the following classical order statistics properties,

$$F_{Y_i}\left(Y_{i,(j)} \mid \theta_i\right) \sim U_{i,(j)}, \quad F_{Y_i}\left(Y_{i,(j)} \mid \hat{\theta}_i\right) \approx U_{i,(j)},$$
$$F_{Y_{i,(j)}}\left(Y_{i,(j)} \mid \theta_i\right) \sim U_{i,j}, \quad F_{Y_{i,(j)}}\left(Y_{i,(j)} \mid \hat{\theta}_i\right) \approx U_{i,j},$$

where $\approx$ means approximately distributed, $\sim$ means distributed as, $U_{i,j} \sim U(0,1)$, and $U_{i,(j)} \sim \text{Beta}(j, n_i + 1 - j)$ and the quality of the approximation in the right hand side depends upon the estimator $\hat{\theta}_i$ and the sample size.

We now connect the order statistics to the underlying talent distribution that comes from a population with $N_i \geq n_i$ observations when F_X is known. This connection is established with

$$F^{-1}_{X_{i,(r(N_i,n_i,j))}}\left(F_{Y_{i,(j)}}\left(Y_{i,(j)} \mid \theta_i\right)\right) \sim F^{-1}_{X_{i,(r(N_i,n_i,j))}}(U_{i,j}) = X_{i,(r(N_i,n_i,j))}. \tag{1}$$

We estimate the above with

$$F^{-1}_{X_{i,(r(N_i,n_i,j))}}\left(F_{Y_{i,(j)}}\left(Y_{i,(j)} \mid \hat{\theta}_i\right)\right) \approx F^{-1}_{X_{i,(r(N_i,n_i,j))}}\left(U_{i,j}\right) = \widehat{X}_{i,(r(N_i,n_i,j))}. \tag{2}$$

Similarly, we can obtain Y from X,

$$F^{-1}_{Y_{i,(j)}}\left(F_{X_{i,(r(N_i,n_i,j))}}\left(X_{i,(r(N_i,n_i,j))}\right)|\theta_i\right) = Y_{i,(j)}, \tag{3}$$

where we estimate the above with

$$F^{-1}_{Y_{i,(j)}}\left(F_{X_{i,(r(N_i,n_i,j))}}\left(X_{i,(r(N_i,n_i,j))}\right)|\hat{\theta}_i\right) = \widehat{Y}_{i,(j)}.$$

Not sure where cross-contextual comparisons are needed. We can use the Full House Model to obtain estimated statistics in another context. Consider the the pair $(Y_{i,(j)}, X_{i,(r(N_i,n_i,j))})$ in the ith population. We first put the aptitude value $X_{i,(r(N_i,n_i,j))}$ obtained by (2) in a new population m and reverse the process to obtain the hypothetical statistics in system m, which we will denote as $Y_{i,j,m}$. More formally, $Y_{i,j,m}$ are estimated as follows in the parametric context,

$$\widehat{Y}_{i,j,m} = F^{-1}_{Y_m}\left(F_{X_{m,(r(N_m,n_m,l_{i,j,m}))}}\left(\widehat{X}_{i,(r(N_i,n_i,j))}\right)|\hat{\theta}_m\right), \tag{4}$$

where $l_{i,j,m}$ is the rank of $\widehat{X}_{i,(r(N_i,n_i,j))}$ among the values

$$\left\{\widehat{X}_{i,(r(N_i,n_i,j))}, \widehat{X}_{m,(r(N_m,n_m,t))} : t = 1,\ldots,n_m\right\}.$$

The estimates (4) correspond to

$$Y_{i,j,m} = F^{-1}_{Y_m}\left(F_{X_{m,(r(N_m,n_m,l_{i,j,m}))}}\left(X_{i,(r(N_i,n_i,j))}\right) \mid \theta_m\right), \tag{5}$$

where $l_{i,j,m}$ is the rank of $X_{i,(r(N_i,n_i,j))}$ among the values

$$\left\{X_{i,(r(N_i,n_i,j))}, X_{m,(r(N_m,n_m,t))} : t = 1,\ldots,n_m\right\},$$

and $X_{i,(r(N_i,n_i,j))}$ is computed via (1).

Paradoxes

Consider two populations $i = 1, 2$ with sizes N_1 and N_2, and systems of sizes n_1 and n_2. We define a paradox (or a Gould paradox) when

$$X_{1,(N_1)} > X_{2,(N_2)} \quad \text{and} \quad Y_{1,(n_1)} < Y_{2,(n_2)}$$

both occur. A central interest is evaluating the probability of such a paradox,

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right). \tag{6}$$

As a simple illustrative example of some mechanics concerning paradoxes, suppose that $Y_{1,j} < Y_{2,k}$ a.s. Then (6) is equivalent to $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right)$, and we have $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right) \to 1$ as $N_1 \to \infty$. We motivate a convenient representation of (6) in the following proposition.

Proposition 0.1. *Consider two populations $i = 1, 2$ with sizes N_1 and N_2, and systems of sizes n_1 and n_2. Suppose that the system inclusion mechanism has the property that $g(X_{i,(N_i)}) = 1$, and that the Full House Model is of the form*

$$\begin{aligned} Y_{i,j} &\sim F_Y(y), \qquad i = 1, 2, \qquad j = 1, \ldots, n_1, \\ X_{i,j} &\sim F_X(x), \qquad i = 1, 2, \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $F_Y^{-1}(y)$ and $F_X^{-1}(x)$ exist. Then,

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < h_{1,2}\left(X_{2,(N_2)}\right)\right), \tag{7}$$

where $h_{i,j}(\bullet) = (f_i^{-1} \circ f_j)(\bullet)$, $f_i(\bullet) = F^{-1}_{Y_{i,(n_i)}}\left(F_{X_{i,(N_i)}}(\bullet)\right)$, and both f and h exist.

Proof. If f and h exist, then we have

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &\quad = \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, f_1(X_{1,(N_1)}) < f_2(X_{2,(N_2)})\right) \\ &\quad = \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, f_1(X_{1,(N_1)}) < f_2(X_{2,(N_2)})\right) \\ &\quad = \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < h_{1,2}\left(X_{2,(N_2)}\right)\right). \end{aligned}$$

Existance of f_i follows directly from the existance of $F_Y^{-1}(y)$, and existance of $h_{i,j}$ follows directly from the existance of f_i and the existance of $F_X^{-1}(x)$. □

Lemma 0.2. *Suppose that the system inclusion mechanism has the property that $g(X_{i,(N_i)}) = 1$, and that the Full House Model is of the form*

$$\begin{aligned} Y_{i,j} &\sim F_Y(y), \qquad i = 1, 2, \qquad j = 1, \ldots, n_1, \\ X_{i,j} &\sim F_X(x), \qquad i = 1, 2, \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $F_Y^{-1}(y)$ and $F_X^{-1}(x)$ exist. Then,

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \frac{n_2}{n_1 + n_2} - \frac{N_2}{N_1 + N_2}.$$

Proof. We can use the convenient representation of (7) since the conditions of Proposition 0.1

hold. Thus, we have

$$
\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < h_{1,2}\left(X_{2,(N_2)}\right)\right) \\
&= \int_{-\infty}^{+\infty} \left[\int_{x}^{h_{1,2}(x)} f_{X_{1,(N_1)}}(y)\, dy\right] f_{X_{2,(N_2)}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left[\int_{x}^{h_{1,2}(x)} N_1 (F_X(y))^{N_1-1} f_X(y) dy\right] N_2 (F_X(x))^{N_2-1} f_X(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left[\int_{x}^{F_X^{-1}(F_X(x)^{\frac{N_2}{N_1}\frac{n_1}{n_2}})} N_1 (F_X(y))^{N_1-1} f_X(y) dy\right] N_2 (F_X(x))^{N_2-1} f_X(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left(F_X(x)^{N_2\frac{n_1}{n_2}} - F_X(x)^{N_1}\right) N_2 (F_X(x))^{N_2-1} f_X(x)\, dx \\
&= \int_{0}^{1} t^{\frac{n_1}{n_2}} - t^{\frac{N_1}{N_2}}\, dt \\
&= \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2},
\end{aligned}
$$

where

$$
h_{1,2}(x) = F_X^{-1}((F_Y(F_Y^{-1}(F_X^{\frac{N_2}{n_2}})))^{\frac{n_1}{N_1}})(x) = F_X^{-1}(F_X^{\frac{N_2}{N_1}\frac{n_1}{n_2}})(x).
$$

□

It is interesting to note that the paradox probability $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \frac{n_2}{n_1+n_2}$ as $N_1 \to \infty$ under the setup of Lemma 0.2.

We verify Lemma 0.2 with a Monte Carlo simulation. In each simulation, we first randomly generate two groups of observation: Y_1 and Y_2 from the standard normal distribution, and the number of each group is given by n_1 and n_2. Based on our setting, we assume the latent aptitudes follow the Pareto distribution with $\alpha = 1.16$. Then we obtain the latent aptitudes for each group and check whether the paradox happens. We repeat this process 1000 times and record the frequency of the paradox. 1000 Monte Carlo iterations of our simulation are performed and we compute the mean of the frequency.

	$n_1 = 400, n_2 = 600, N_1 = 1e6, N_2 = 1e6$	$n_1 = 300, n_2 = 700, N_1 = 1e6, N_2 = 1e6$
Prob from lemma or ties	0.1	0.2
Prob from simulation	0.10084	0.200309

Table 1: Simulation result to validate the lemma about probability of paradox

Definition 0.3. *We will say that Y_2 is first order ε-weakly stochastically dominant over Y_1 if, for $\varepsilon > 0$, $\mathrm{P}(Y_2 \le y) \le \mathrm{P}(Y_1 \le y) \le \mathrm{P}(Y_2 \le y) + \varepsilon$ for all y and $\mathrm{P}(Y_2 \le y) < \mathrm{P}(Y_1 \le y) \le \mathrm{P}(Y_2 \le y) + \varepsilon$ for some y.*

Lemma 0.4. *Suppose that the system inclusion mechanism has the property that* $g(X_{i,(N_i)}, \boldsymbol{X}_{i,-(N_i)}) = 1$*, and that the Full House Model is of the form*

$$\begin{aligned} Y_{i,j} &\sim F_{Y_i}(y), \qquad i = 1, 2, \qquad j = 1, \ldots, n_1, \\ X_{i,j} &\sim F_X(x), \qquad i = 1, 2, \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $F_{Y_i}^{-1}(y)$ *and* $F_X^{-1}(x)$ *exist. Further suppose that* Y_2 *is first order* ε*-weakly stochastically dominant over* Y_1 *for some* $\varepsilon > 0$*. Then,*

$$\begin{aligned} \frac{n_2}{n_1 + n_2} - \frac{N_2}{N_1 + N_2} &\leq \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &\leq \frac{n_2}{n_1 + n_2} + \varepsilon \frac{n_1 n_2}{n_1 + n_2 - 1} - \frac{N_2}{N_1 + N_2} + o(\varepsilon). \end{aligned} \tag{8}$$

Proof. First, observe that the assumption of first order ε-weakly stochastically dominant gives

$$\left(F_{Y_{2,(n_2)}}(y)\right)^{n_1/n_2} \leq \left(F_{Y_1}(y)\right)^{n_1} \leq \left(F_{Y_2}(y) + \varepsilon\right)^{n_1} \leq \left(\left(F_{Y_{2,(n_2)}}(y)\right)^{1/n_2} + \varepsilon\right)^{n_1},$$

and this implies that

$$F_X^{-1}\left(F_X^{\frac{N_2}{N_1}\frac{n_1}{n_2}}(X_{2,(N_2)})\right) \leq h_{1,2}(X_{2,(N_2)}) \leq F_{X_{1,(N_1)}}^{-1}\left(\left(\left(F_{X_{2,(N_2)}}(X_{2,(N_2)})\right)^{1/n_2} + \varepsilon\right)^{n_1}\right).$$

In combination with Lemma 0.2 and the proof of Lemma 0.2, the first inequality yields

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \geq \frac{n_2}{n_1 + n_2} - \frac{N_2}{N_1 + N_2}.$$

This verifies the first inequality of (8). We now verify the second inequality of (8). Now,

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &\quad \leq \int_{-\infty}^{\infty} \left[\int_x^{F_{X_{1,(N_1)}}^{-1}\left(\left(\left(F_{X_{2,(N_2)}}(x)\right)^{1/n_2} + \varepsilon\right)^{n_1}\right)} f_{X_{1,(N_1)}}(y)dy\right] f_{X_{2,(N_2)}}(x)dx \\ &\quad = \int_{-\infty}^{\infty} \left[\left(\left(F_X(x)\right)^{N_2/n_2} + \varepsilon\right)^{n_1} - F_X(x)^{N_1}\right] N_2 F_X(x)^{N_2 - 1} f_X(x)dx \\ &\quad = N_2 \int_{-\infty}^{\infty} \left(\left(F_X(x)\right)^{N_2/n_2} + \varepsilon\right)^{n_1} F_X(x)^{N_2 - 1} f_X(x)dx - \frac{N_2}{N_1 + N_2}. \end{aligned}$$

A first-order Taylor expansion of $\left(\left(F_X(x)\right)^{N_2/n_2} + \varepsilon\right)^{n_1}$ gives,

$$\left(\left(F_X(x)\right)^{N_2/n_2} + \varepsilon\right)^{n_1} = \left(F_X(x)\right)^{N_2 n_1/n_2} + \varepsilon n_1 \left(\left(F_X(x)\right)^{N_2/n_2}\right)^{n_1 - 1} + o(\varepsilon).$$

Thus,

$$
\begin{aligned}
&N_2 \int_{-\infty}^{\infty} \left((F_X(x))^{N_2/n_2} + \varepsilon \right)^{n_1} F_X(x)^{N_2-1} f_X(x) dx \\
&\quad = N_2 \int_{-\infty}^{\infty} \left[(F_X(x))^{N_2 n_1/n_2} + \varepsilon n_1 \left((F_X(x))^{N_2/n_2} \right)^{n_1-1} \right] F_X(x)^{N_2-1} f_X(x) dx + o(\varepsilon) \\
&\quad = \frac{n_2}{n_1+n_2} + \varepsilon n_1 N_2 \int_{-\infty}^{\infty} F_X(x)^{N_2((n_1-1)/n_2+1)-1} f_X(x) dx + o(\varepsilon) \\
&\quad = \frac{n_2}{n_1+n_2} + \varepsilon \frac{n_1 n_2}{n_1+n_2-1} + o(\varepsilon).
\end{aligned}
$$

Putting it all together yields the second inequality of (8). □

Example: Uniform distributions. In this example we demonstrate a case where Lemma 0.4 applies, but we can explicitly calculate the paradox probability. Suppose that the system inclusion mechanism has the property that $g(X_{i,(N_i)}, \mathbf{X}_{i,-(N_i)}) = 1$, and we will consider the following Full House Model for two systems,

$$
\begin{aligned}
Y_{1,j} &\sim U(0,1), \qquad j = 1, \ldots, n_1, \\
Y_{2,j} &\sim U(0, 1+\delta), \qquad j = 1, \ldots, n_2, \\
X_{i,j} &\sim U(0,1), \qquad j = 1, \ldots, N_i,
\end{aligned}
$$

where $\delta > 0$. We can pick $\varepsilon = \delta/(1+\delta)$ so that Y_2 is first order ε-weakly stochastically dominant over Y_1 with this choice of ε, and the conditions of Lemma 0.4 are satisfied. However, our setup allows for direct calculation of the paradox probability. From Proposition 0.1 and a bit of manipulation we have that

$$
\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} \right) = \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < h_{1,2}\left(X_{2,(N_2)} \right) \right) \\
&= \int_0^1 \int_x^{h_{1,2}(x)} f_{X_{1,(N_1)}}(y)\, dy\, f_{X_{2,(N_2)}}(x)\, dx \\
&= \int_0^1 \int_x^{h_{1,2}(x)} N_1 y^{N_1-1}\, dy\, N_2 x^{N_2-1}\, dx \\
&= \int_0^1 \int_x^{((1+\delta)x^{\frac{N_2}{n_2}})^{\frac{n_1}{N_1}}} N_1 y^{N_1-1}\, dy\, N_2 x^{N_2-1}\, dx \\
&= \int_0^1 \left(((1+\delta)x^{\frac{N_2}{n_2}})^{n_1} - x^{N_1} \right) N_2 x^{N_2-1}\, dx \\
&= \frac{n_2}{n_1+n_2}(1+\delta)^{n_1} - \frac{N_2}{N_1+N_2}.
\end{aligned}
$$

Here we see that the probability of paradox is 1 for all $\delta \geq [2(n_1/n_2+1)]^{1/n_1} - 1$. When $n_1 \gg n_2$, then $[2(n_1/n_2+1)]^{1/n_1} - 1 \approx n_1^{n_1} - 1$ and we see that the probability of paradox is 1 for extremely small values of δ.

Note

Let $Y_1 \sim N(0, \sigma)$ and $Y_2 \sim N(2\sigma, \sigma)$. Let

$$a(y) = \frac{F_{Y_1}(y) - F_{Y_2}(y)}{2\sigma}.$$

Then $a(y)$ is the density of a normal distribution with mean σ and standard deviation $1/\left(\sqrt{2\pi}a(\sigma)\right)$. *This needs to be verified.*